

Paul He

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Education

Nanyang Technological University , PhD in Computer Science <i>Supervisor: Prof. Kwok Yan Lam (College of Computing and Data Science)</i>	Jan 2026 – Dec 2030 Singapore
• Incoming PhD student with full research scholarship University of Toronto , BSc in Computer Science (Honours) <i>Supervisors: Prof. Gerald Penn, Prof. Zhijing Jin</i>	Sep 2021 – May 2025 Toronto, Canada
• Graduated with High Distinction ETH Zürich , Exchange Year in Computer Science • Completed graduate-level coursework in NLP and Machine Learning	Sep 2023 – Aug 2024

Work Experience

Amazon Web Services (AWS) <i>Applied Scientist II, CloudWatch Logs</i>	Sep 2025 – Dec 2025 Tübingen, Germany
• Conducted open-ended research on reliability and evaluation of large language models in production settings. • Implemented novel evaluation and reliability methods as production-quality code in internal pipelines. • Contributed research directions that developed into ongoing manuscripts and collaborations.	

Research Experience

Causal Reasoning Evaluation for Language Models <i>Supervisor: Prof. Zhijing Jin (Jinesis AI Lab)</i>	Jan 2025 – Aug 2025 Vector Institute
• Designed a sound & complete symbolic verification metric to uncover correctness missed by standard evaluations.	
Polynomial-Time CCG Parsing <i>Supervisor: Prof. Gerald Penn</i>	Sep 2024 – Aug 2025 University of Toronto
• Conducted large-scale multilingual experiments identifying empirical complexity thresholds where symbolic parsing outperforms naive baselines.	
Unsupervised Prompt Optimization for RAG using PMI <i>Supervisors: Tianyu Liu, Prof. Ryan Cotterell (Rycolab)</i>	Jun 2024 – Sep 2024 ETH Zürich
• Developed an unsupervised PMI-based document reordering method, improving QA accuracy by up to 4%.	

Publications

- [1] **Uncovering Hidden Correctness in LLM Causal Reasoning via Symbolic Verification**
Paul He, Yinya Huang, Mrinmaya Sachan, Zhijing Jin. *EACL 2026 (Main)*. Also accepted to *NeurIPS 2025 CauScien Workshop (non-archival)*.
- [2] **CCG Revisited: A Multilingual Empirical Study of the Kuhlmann–Satta Algorithm**
Paul He, Gerald Penn. *IWPT 2025*. aclanthology.org/2025.iwpt-1.3
- [3] **Pointwise Mutual Information as a Performance Gauge for Retrieval-Augmented Generation**
Tianyu Liu*, Jirui Qi*, Paul He, Arianna Bisazza, Mrinmaya Sachan, Ryan Cotterell. *NAACL 2025 (Main)*. aclanthology.org/2025.naacl-long.78

Preprints & Under Review

- [4] **Dialogue as Uncertainty Reduction: A Judge-Free Metric for Multi-Turn Dialogue Evaluation**
Paul He, Shiva Kasiviswanathan‡, Dominik Janzing‡. *Manuscript in preparation for submission to ARR (Jan 2026)*.
- [5] **Foundations of Global Consistency Checking with Noisy LLM Oracles**
Paul He, Elke Kirschbaum, Shiva Kasiviswanathan. *Under review at ARR*.
- * Equal contribution. ‡ Equal supervision.

Skills

Programming Languages: Python (research & production), C, Java, SQL

Languages: English (fluent), Mandarin (proficient), German (proficient)